

Digital technologies and sustainability accounting, reporting and assurance: framework and research opportunities

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Abstract

Purpose – The purpose of this special issue (SI) lead paper is to examine the role of digital technologies in sustainability accounting, reporting and assurance; review the articles in this SI; and identify future research directions.

Design/methodology/approach – This paper draws on academic literature to explore the role of digital technologies in sustainability accounting, reporting and assurance.

Findings – Digital technologies can assist in setting disclosure scope and objectives (e-taxonomies); undertaking materiality assessments (digital communication, artificial intelligence and big data analytics); collecting and analysing data (Internet of Things and radio frequency identification); potentially revolutionizing sustainability assurance practices; and disseminating disclosures (interactive reports). Papers in this SI examine a range of digital technologies and issues. Future research can examine the impact of Industry 4.0, barriers to adoption (and overcoming these), use of alternative theories and methodologies, critical examination of the benefits of digitalization and impact of digital technologies on the sustainability reporting and sustainability assurance processes.

Research limitations/implications – This paper seeks to set the agenda for contemporary and future research on sustainability accounting, reporting and assurance and digital technologies.

Practical implications – The findings from this study have implications for regulators, standard setters and practitioners (including sustainability reporting managers and sustainability assurance providers), highlighting the impact of digital technologies on sustainability accounting, reporting and assurance.

Social implications – Sustainability accounting, reporting and assurance concepts, techniques and practices play a key role in promoting transparency and organizational accountability to stakeholders on sustainability.



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Understanding how digital technologies can be leveraged to enhance sustainability accounting, reporting and assurance offers benefits to stakeholders and society.

Originality/value – This research offers a holistic view of the entire spectrum of sustainability accounting, reporting and assurance concepts, technologies and practices and the potential role digital technologies can play in them. It also offers insights for future research in a contemporary area.

Keywords Sustainability reporting, Sustainability accounting, Digital technologies, Sustainability assurance, United Nations Sustainable Development Goals (UN SDGs), Disruptive transformation

Paper type Research paper

1. Introduction

Digital technologies are profoundly impacting the world we live in (De Villiers *et al.*, 2021). The constant flow of innovative technologies is driving disruptive transformations across all aspects of life (Lodhia, 2018; Lodhia and Stone, 2017). Digital technologies impact individuals, organizations and society, shaping how we think, behave and operate. While multiple types of digital technologies exist, our focus is on contemporary digital technologies such as big data analytics, artificial intelligence (AI) and machine learning, blockchain and cryptocurrency, social media communication, cloud computing and the Internet of things (IoTs). Rapid advancements in these technologies provide numerous applications for the field of accounting and a need for academic investigation into their use. In this paper, we explore the impact of digital technologies on sustainability accounting, reporting and assurance.

The demand for sustainable development is fundamental change shaping contemporary society. Sustainability is no longer perceived as a desirable but optional element of society; rather it is acknowledged as central to our survival. Sustainability accounting, reporting and assurance emerged as critical concepts, techniques and practices to promote transparency and accountability on organizations social, economic and environmental performance and in doing so support societal aspirations for sustainable growth and development (Farooq and De Villiers, 2019a; Lodhia, 2018). This involves collecting, analysing and reporting on organizations sustainability impacts (Farooq and De Villiers, 2019a; Kaur and Lodhia, 2018; Lodhia, 2024). Furthermore, sustainability assurance emerged to provide assurance over the credibility of organizations sustainability disclosures (Farooq and De Villiers, 2019b, 2020; Farooq *et al.*, 2024a, 2024b).

The rapid advancements in digital technologies coupled with the increasing attention given to sustainability accounting, reporting and assurance by practitioners, regulators and policymakers, provides an opportunity for this special issue to advance limited extant literature in this contemporary area. Lodhia *et al.* (2020) explored the use of social media technologies for sustainability reporting highlighting that such technologies are changing the dynamics of sustainability reporting. De Villiers *et al.* (2021) examine the potential role of the IoTs to facilitate recording data and tracking progress on the United Nations Sustainable Development Goals. Other researchers offer a different perspective on the potential impact of digital technologies. For example, the managers interviewed by Tiwari and Khan (2020) were less than optimistic about the transformative potential of contemporary digital technologies. In their literature review, Xu *et al.* (2019) examine the impact of blockchain on organizations. They shed light on the transformative potential of blockchain technologies and identify several avenues for future research. In a reflective study, Burritt and Christ (2016) argue in favour of allocating greater academic effort to understanding these unexplored phenomena. Thus, the important and contemporary nature of this topic requires greater research, and this special issue attempts to address this gap in our knowledge base by

contributing to an improved understanding of the impact of digital technologies on sustainability accounting, reporting and assurance.

The remainder of this paper is organized as follows. We relate digital technologies to sustainability accounting, sustainability reporting and sustainability assurance in Section 2. This is followed by an overview of the papers in this special issue in Section 3. We highlight potential future areas of research in Section 4. We finish the paper by offering some concluding comments in Section 5.

2. Impact of contemporary digital technologies on sustainability accounting, reporting and assurance

The rapid advancement of digital technologies is transforming sustainability accounting, reporting and assurance practices. Technologies such as AI, big data analytics, blockchain, cloud computing and the IoT are revolutionizing environmental, social and governance (ESG) data collection, analysis and reporting, while robotic process automation (RPA), 3D visualization and social media disclosure further streamline processes, enhance stakeholder engagement and strengthen assurance mechanisms. A key driver of this shift is Industry 4.0, which integrates IoT, AI, big data and cloud computing to enable real-time tracking of energy consumption, carbon emissions and waste, thereby ensuring alignment with global sustainability standards.

Beyond accounting and reporting, digitalization is redefining sustainability assurance. Blockchain enhances data security and traceability, mitigating greenwashing risks, while AI-driven fraud detection improves risk assessment. IoT-enabled auditing further facilitates remote ESG data verification. However, challenges such as cybersecurity risks, data privacy concerns, regulatory gaps and the potential misuse of digital tools persist. [Table 1](#) provides an overview of key digital technologies, their applications, benefits and challenges in sustainability accounting, reporting and assurance, with subsequent subsections explaining their impacts and the research gaps in their adoption.

2.1 Contemporary digital technologies and sustainability accounting

The role of digital technologies in sustainability accounting represents an emerging field of enquiry that is receiving increasing interest from scholars and practitioners ([Lodhia, Kaur, and Stone, 2020](#)). The advancement of these technologies has created a new environment for data collection and processing, offering unprecedented opportunities to support sustainability goals. Through the digitalization of company operations, vast amounts of data (or Big Data) can be harnessed to facilitate sustainability accounting systems, thereby enabling informed decision-making and fostering more sustainable business processes ([Klymenko et al., 2021](#)). Industry 4.0, widely regarded as a transformative framework, plays a pivotal role in helping companies achieve these sustainability objectives ([Petcu, Sobolevschi-David and Curea, 2024](#)).

Industry 4.0 refers to the fourth industrial revolution and represents integration of digital technologies such as IoT, AI, big data analytics and advanced automation into manufacturing and other industries (see [Table 1](#)). In the context of sustainability accounting, reporting and assurance, industry 4.0 technologies can play a critical role in accomplishing transparency, accuracy and efficiency. IoT sensors embedded in machines or production processes can collect real-time environmental data, such as energy consumption, water usage, carbon emissions and waste production. Such data can be fed into sustainability reports, ensuring that organizations track and measure their environmental impact accurately ([Petcu et al., 2024](#)). Big data analytics' can also be used to assess the life cycle of products and services, helping businesses understand their broader environmental footprint.

Table 1. Digital technology types, uses, benefits and challenges, in sustainability accounting, reporting and assurance

Technology types	Uses			Limitations and challenges
	Sustainability accounting	Sustainability reporting	Sustainability assurance	
Cloud computing	Centralized ESG data storage, supports real-time accounting and integration across supply chains	Automates ESG reporting (e.g. XBRL for regulatory compliance), enables data-driven disclosures	Facilitates remote audits, allowing assurance providers to verify ESG data from anywhere	Cybersecurity risks, concerns over data sovereignty and privacy
Big data analytics	Processes large ESG datasets for sustainability performance tracking	Enhances dynamic, real-time sustainability disclosures, improves materiality assessment	Enables risk assessments, fraud detection and assurance quality improvement	High costs, requires data governance frameworks
Artificial intelligence (AI) and NLP	Automates ESG accounting, detects anomalies and enhances decision-making	AI-powered tools generate sustainability disclosures, ensuring consistency	It can detect greenwashing and inconsistencies but can also be misused for AI-generated misleading reports	Risk of bias, misinformation and regulatory concerns (e.g. ChatGPT-generated sustainability claims)
Internet of Things (IoT)	Automates real-time ESG data collection (e.g. carbon emissions, energy consumption)	Provides real-time ESG metrics for integrated reporting	Improves auditability and verifiability of sustainability data by reducing reliance on manual processes	Interoperability issues across global operations, cybersecurity vulnerabilities
Blockchain technology	Recognized for its ability to provide an immutable and transparent ledger, reducing the risk of fraud and enhancing the reliability of financial statements	Enhances transparency and traceability of ESG disclosures, reducing greenwashing risks	Creates verifiable audit trails, improving credibility in assurance engagements	Scalability issues, regulatory acceptance still evolving

(continued)

Table 1. Continued

Technology types	Uses			Limitations and challenges
	Sustainability accounting	Sustainability reporting	Sustainability assurance	
3D Game Engine Technology and Virtual Reality (VR)	Improves stakeholder engagement by creating interactive ESG data models	Enhances visual storytelling in sustainability reports	Can simulate real-world assurance scenarios, improving training for assurance professionals	Developing and integrating 3D game engines and VR into sustainability reporting and assurance requires significant investment in software, hardware and skilled personnel. Costs may be prohibitive for small and medium-sized organizations
Social media disclosure (Facebook, X, LinkedIn)	Supports real-time financial and ESG disclosures, fostering public engagement	Enhances transparency through interactive sustainability updates and supports stakeholder engagement in the reporting process	Enables stakeholder feedback loops, potentially improving assurance credibility	Risk of misinformation, reputational damage and greenwashing. Negative public reaction to social media posts is a challenge and can exert additional pressure on companies
Robotic process automation (RPA)	Automates ESG data entry, taxonomies and compliance reporting. An RPA bot can scan and extract key information (e.g. invoice number, date, amount, supplier details) from emails or PDFs using optical character recognition (OCR)	RPA enhances sustainability reporting by reducing human error in data extraction, ensuring accuracy and consistency and automating compliance with frameworks like GRI and SASB. It streamlines carbon footprint tracking, enables real-time monitoring and improves overall efficiency and transparency in reporting	Improves audit efficiency by automating compliance checks and risk assessments	Limited adaptability for unstructured ESG data

(continued)

Table 1. Continued

Technology types	Uses		Benefits	Limitations and challenges
	Sustainability accounting	Sustainability reporting		
Data visualization and dashboards	Enhances financial and ESG data clarity through interactive dashboards	Improves readability of sustainability reports with visual storytelling	Makes assurance reports more accessible to nonexperts via graphs, charts and interactive tools	Risk of misleading visualizations being used for greenwashing

Source(s): Authors' own work

Source(s): Authors' own work

Industry 4.0 tools can automate and streamline sustainability accounting by continuously tracking resources used (energy, water, raw materials) and waste produced in real time. This enables organizations to manage their environmental impacts more efficiently and reduce costs associated with inefficiencies. Industry 4.0 technologies enable seamless integration with ESG reporting frameworks. Organizations would be able to provide accurate and timely data in alignment with global reporting standards such as the Global Reporting Initiatives (GRI) standards and the more recent S1 and S2 issued by the International Sustainability Standards Board (ISSB) of the International Finance Reporting Standards Foundation (IFRS Foundation).

Despite its potential, the adoption of Industry 4.0 technologies for sustainability accounting remains underexplored. For instance, [Tiwari and Khan \(2020\)](#) highlight the limited empirical evidence in this domain, underscoring the need for further research and practical implementation. Sustainability itself, as defined by the United Nations World Commission on Environment and Development ([UNWCED, 1987](#)), aims to meet present needs without compromising the ability of future generations to meet theirs. However, achieving this ideal is hindered by organizational policies and practices that often seek to further personal interests while ignoring social, economic and environmental impacts on society and the planet. [Burritt and Christ \(2016\)](#) note that many companies lack the technological infrastructure required to collect timely and accurate data, which can undermine the credibility of environmental efforts and expose firms to accusations of greenwashing. While financial accounting has evolved over several centuries, providing a detailed view of companies, financial operations, accounting systems for tracking corporate sustainability performance remain relatively under development ([Hourneaux et al., 2018](#)).

Sustainability accounting data serve several critical purposes. They enable organizations to demonstrate their progress in addressing social, economic and environmental goals to stakeholders, enhancing transparency and organizational accountability. Internally, these data inform decision-making processes, allowing organizations to optimize operations in line with sustainability goals, such as maintenance, reuse, remanufacturing and recycling. In addition, the data provide insights for reducing material consumption, greenhouse gas emissions and waste ([Burritt and Christ, 2016](#)). Beyond operational improvements, sustainability accounting data also play a strategic role, guiding organizations towards the adoption of sustainable business models.

Digitalization and Industry 4.0 is receiving growing attention in supply chains among both practitioners and academics. Digitalization is the driver of Industry 4.0 and is shaping the digitalization of a range of industries, particularly those in the primary and tertiary sectors of the economy. The large volumes of data generated through digitalization can theoretically be harnessed to facilitate sustainability accounting at zero cost ([Klymenko et al., 2021](#)). Digital technologies provide the information needed to create iterative and restorative systems, enabling organizations to move towards sustainable operations and products.

By coupling IoT-enabled innovation with sustainability principles, organizations may identify new business models. Through digitalization, it is possible for organizations to improve and optimize capacity utilization, resource efficiency and inventory management operating, and apply technology for predictive maintenance ([Klymenko et al., 2021](#)). Furthermore, digital technologies can potentially facilitate decision-making during uncertainty.

[Burritt and Christ \(2016\)](#) note that the focus on the development of Industry 4.0 has been on “reducing errors, improving product quality, freeing humans from menial and/or dangerous tasks and providing customers with the products they desire at times when they desire them” (p. 24). They note that resource efficiency is underemphasized and argue that

the broader concept of corporate sustainability incorporation into Industry 4.0 is underdeveloped. To address this issue, [Burritt and Christ \(2016\)](#) examine how corporate sustainability through environmental accounting – might be incorporated into developing a vision for Industry 4.0. They mention four potential improvements in external environmental accounting (P.30): better data quality (timeliness, accuracy, reliability, comparability); reduced opportunity for “greenwash” and “brownwash”; less management discretion regarding measurements; and high credibility of data.

The main motive for deploying digital technologies in sustainability accounting may come from meeting stakeholder (particularly customer and supplier) expectations for product and service quality. Strategic decision-making of companies is mainly supported by economic benefits ([Klymenko et al., 2021](#)), otherwise, companies only engage in sustainability to comply with regulations, or respond to specific customer pressure. The Industrial Internet of Things (IIoT) and big data analytics are at the core of Industry 4.0 ([Kiel et al., 2017](#)). Currently, manufacturing organizations are adopting IIoT and big data systems for addressing gaps in industrial process data collection and analysis. According to [Tiwari and Khan \(2020\)](#), the capability of Industry 4.0 may be viewed as the next giant step of innovation beyond computer aided designs, computer-aided engineering and computer-aided manufacturing. The IIoTs provide time-series data about process performance attributes in real time to big data analytics, which uses AI to determine and fine tune factor variables which shape production effectiveness ([Kiel et al., 2017](#)). Industry 4.0 offers benefits in the areas of automation and operation efficiency as well as in sustainability accounting and reporting. The next section examines role of contemporary digital technologies in sustainability reporting.

2.2 Contemporary digital technologies and sustainability reporting

To examine the potential application of digital technologies in sustainability reporting, we draw on the work of [Farooq and De Villiers \(2019a, 2019b\)](#), who explore the institutionalization of sustainability reporting in organizations. Their work allows researchers to delineate the key stages in the reporting process including key issues sustainability reporting managers (SRMs) must grapple with when preparing sustainability reports. Thus, the framework offers a useful tool to discuss the potential application of digital technologies at each stage of the reporting process.

2.2.1 Setting the scope and objectives of sustainability reporting. The first stage in the reporting process involves SRMs deciding on which standard or combination of standards to use when preparing their sustainability disclosures. A review of the landscape reveals a plethora of standards, from the Global Reporting Initiative’s (GRI) suit of standards, the Greenhouse Gas (GHG) Protocols and the recommendations of the Task Force on Climate Related Financial Disclosures. To complicate matters, the landscape continues to change as old standards are shelved, and new ones introduced. For instance, the International Integrated Reporting Council was launched in 2011, issued a framework on integrated reporting in 2013 and was dissolved 11 years later in 2021 ([Farooq et al., 2024a, 2024b](#)). The European Union (EU) has issued a new Corporate Sustainability Reporting Directive (CSRD) and has announced that it is working on a set of EU specific standards. The choice of standard (or standards) to use may be influenced by government regulations and/or stock exchange listing requirements. The scope and objectives of these standards varies with some focusing on specific aspects of sustainability, while others provide guidance on the full remit of sustainability issues that reporters may encounter. Furthermore, the GRI offers reporters the option of preparing their disclosures either in accordance with (full application) or with reference (partial application) to their standards.

The use of taxonomies, particularly e-taxonomies, may present a solution (or a partial solution) to this challenge. The EU taxonomy for sustainable economic activities is one such example. The EU describes its taxonomy as a “classification system that defines criteria for economic activities that are aligned with a net zero trajectory by 2050 and the broader environmental goals other than climate” (European Commission n.d.). Companies that must report under the EU’s new CSRD need to identify activities that are covered in the EU’s taxonomy. Companies exempt from reporting under the CSRD can still choose to voluntarily disclose this information to their stakeholders (European Commission, n.d.). The European Commission avoids using terms like standardization, consistency or comparability when describing its taxonomy, which is instead presented as a tool to promote transparency by creating “a common language and a clear definition of what is sustainable” (European Union, 2025). European Commission’s business case for the taxonomy is potential access to sustainable finance, while the normative argument is that it prevents greenwashing. Furthermore, while the taxonomy is not digitalized it is arguably the first step towards digitalization which can only be achieved when there is some degree of consensus (i.e. a common language) for sustainability reporting. However, taxonomies (and e-taxonomies), whether intentionally or unintentionally, will introduce some degree of standardization into what is arguably a highly complex and dynamic field, and at a time when our knowledge of sustainability reporting (and sustainability) is still in its infancy and evolving. In such a situation, innovation and experimentation may be preferred over introducing a common language.

2.2.2 Materiality assessment and stakeholder engagement. The materiality assessment is perhaps the most important stage in reporting process, as the outcome of the assessment dictates which topics/issues will be reported on in the sustainability report (Farooq and De Villiers, 2019a, 2019b). The exercise involves identifying and prioritizing issues that are significant/material for both the reporter and its stakeholders (Farooq et al., 2021a). The nondisclosure of material issues results in the publication of unbalanced reports that typically cover good news while failing to discuss material bad news (otherwise known as greenwashing). Recently the concept of double materiality, another example of the evolutionary nature of the field, has gained popularity. Double materiality involves identifying and reporting on how an organization’s operations and activities are impacting society, economy and the environment, as well as how sustainability related issues (e.g. climate change and global warming) are impacting the reporting entity (Global Reporting Initiative, 2025). The first is the more commonly accepted view of materiality, while the second is a more recent edition and explores the financial implications of issues such as climate change. Stakeholder engagement is a key component of the materiality assessment. SRMs identify and engage with stakeholders to understand issues which stakeholders perceive as important. Once key issues are identified managers can identify managers or content owners who can contribute data and information on specific issues.

However, studies find that the quality of corporate materiality assessment require significant improvement (Farooq et al., 2021b). Digital technologies can assist SRMs in several ways during the material assessment. Firstly, digital technologies allow SRMs to engaging with stakeholders. This is particularly important for large geographically dispersed entities. Using digital technologies SRMs can execute online surveys and undertake telephone and video interviews (De Villiers et al., 2024; Farooq and De Villiers, 2018). Secondly, SRMs can use AI and big data analytics to track key issues trending on the internet and analyse discussions on social media platforms and blogs. Digital tools allow SRMs to sift through large amounts of data with relative speed and analyse metrics to understand stakeholder discussions and gauge sentiments (see Table 1).

However, research shows that the use of the internet by listed German companies was restricted primarily to disseminating electronic copies of sustainability reports as this was seen to be a cheaper and more efficient option than posting paper-based copies (Herzig and Godeman, 2010). Manetti and Bellucci (2016) examine corporate use of social media platforms as a tool to engage in dialogue with stakeholders on sustainability related issues. They found that only a small minority of reporting entities leveraged social media and when it was used the level/intensity of interactions was limited. They conclude that these technologies have limited ability to support transparency and accountability. Similar results were reported by Lodhia *et al.* (2020), who found that the use of social media platforms for disseminating sustainability information are not common amongst listed Australian entities. Furthermore, these technologies are often used to support the legitimacy of the reporting entity by controlling the narrative and less to promote genuine dialogue and corporate accountability to stakeholders (Lodhia *et al.*, 2020).

However, managers will need to leverage traditional modes of engagement (face-to-face or telephone communication) to reach out to stakeholders in local communities based in far flung and developing areas where internet speeds and access is limited or nonexistent. These stakeholders will often represent the poorest and most affected sections of society and whose views and concerns are the most important for reporters committed to sustainable development.

2.2.3 Measuring performance; data collection and analysis. Measuring a reporting entity's sustainability performance is challenging as sustainability related impacts are often complex, inter-linked, ill-defined and with far-reaching long-term implications. As a result, many reporters will avoid discussing impacts and instead focus primarily on outputs which are relatively easier to measure. This is one of the reasons why some material issues may go unreported leading to the publication of sustainability reports that lack balance. To further complicate matters, the metrics used to measure and report on sustainability performance continue to change as our knowledge improves but also as various parties lobby for control over this space (e.g. the accounting professions Integrated Reporting project). Currently, many SRMs must design templates for content owners to populate, thereby allowing for the collection of data in format suitable for subsequent analysis (Farooq and De Villiers, 2019a, 2019b).

SRMs may be able to secure some of their sustainability data and information (or sustainability content) needs from existing sources within the reporting entity (Farooq and De Villiers, 2019a, 2019b). For example, many organizations will already be tracking their performance in areas such as health and safety and power/energy consumption. However, in many other areas will require introduction of systems and processes to track performance and to analyse the numbers. Digital technologies including the IoT coupled with real-time data collection and predictive analytics using big data analytics, artificial intelligence and cloud manufacturing can assist SRMs in reporting on material issues in line with the GRI standards (Tiwari and Khan, 2020, p. 120783). For example, organizations can use Radio Frequency Identification (RFID) or Quick Response (QR) codes to collect large volumes of data.

2.2.4 Dissemination. The dissemination of sustainability reports is typically through hard copy or digital/PDF file formats. The latter option is often preferred as it reduces costs and improves reach. However, SRMs often complain that no one is reading their sustainability reports in which they invested considerable time and effort in putting together (Farooq and De Villiers, 2019a, 2019b). Low readership rates may be due to a combination of factors including; challenges in understanding issues of a complex and technical/scientific nature; the large volume of data and information (addressed at least partially through the materiality assessment); the publication of reports that fail to provide information on issues

or locations relevant to stakeholders; and large stakeholders being able to access the information they require through other channels (Farooq and De Villiers, 2019a, 2019b).

The use of interactive reports facilitates disclosure of high-level/summary data with hyperlinks that take stakeholders to underlying data. While this practice has gained some traction amongst reporters, hyperlinks are primarily used to take readers to detailed tables located either within the same document (in appendices) or on websites. Typically, reporters do not hyperlink their reports to the raw data that sits behind the numbers. For example, a reporter may provide summary data on carbon emissions for the year with a hyperlink that takes the reader to a table providing a break-up of carbon emissions across activities and/or locations. However, a stakeholder who lives in a certain location may want more information in relation to a particular location, and this is missing in most reports. Reporters are also making greater use of their websites and social media platforms to communicate with stakeholders on their sustainability performance. These channels of communication may be popular especially amongst certain segments of society and thus effective in opening a dialogue with concerned stakeholders. Furthermore, digital technologies in the form of webcasts and multimedia presentations have the potential to enrich communication with SRM's directly presenting their sustainability report to stakeholders (Lodhia and Stone, 2017). During the presentation, SRMs can answer live questions from globally dispersed stakeholders. Subsequently, the presentation along with the Q&A can be recorded for others to review (read and/or listen) at their own pace.

2.3 Contemporary digital technologies and sustainability assurance

In comparison to sustainability reporting, sustainability assurance is a relatively new phenomena which remains either completely voluntary/unregulated or at best loosely regulated in most jurisdictions across the world [1]. As a consequence, there remains considerable variation in the scope and objectives of sustainability assurance engagements, including differences in what is being assured (data and information or underlying systems and processes or both), who the assurance provider is (accounting or non-accounting) and how the assurance engagement is being undertaken (assurance methodology) (Farooq and De Villiers, 2018). Sustainability assurance providers are typically categorized into accounting and nonaccounting sustainability assurance providers, with significant variation in the scope and depth of their engagements. Farooq and De Villiers (2020) note that many practitioners offer limited assurance over the reliability of the contents of sustainability reports. Thus, assurers avoid providing high levels of assurance over sustainability report reliability and balance, which limits their ability to enhance disclosure credibility and promote transparency and accountability. This challenge is further corroborated by Zaman, Farooq *et al.* (2021), who analysed the assurance quality of sustainability reports issued by the top 100 listed companies in Australia and New Zealand between 2017 and 2019. Their findings revealed consistently poor-quality assurance across both countries, offering little to mitigate stakeholder-agency conflicts and underscoring the need for systemic improvements. Compounding the issue, sustainability assurance is inherently complex due to extensive data requirements and the proliferation of sustainability reporting frameworks and standards (Bakarich, Castonguay and O'Brien, 2020). For example, the new ISSB S1 and S2 standards will have implications for assurance providers.

Reporters are now expected to provide transparent disclosures across a wide range of sustainability related issues, necessitating data collection from diverse sources across their operations and extended supply chains – both upstream and downstream (Bellucci *et al.*, 2022; De Silva *et al.*, 2024). This intricate process demands robust systems for gathering, validating and presenting data, further challenging the sustainability assurance process

(Bakarich, Castonguay and O'Brien, 2020). Furthermore, assurance standards, such as International Audit and Assurance Standard Board's (IAASB's) standard International Standard on Sustainability Assurance (ISSA) 5000, require assurers to issue evidence-based opinions [2]. The convergence of these challenges, coupled with the inadequate quality of current sustainability assurance practices, underscores the urgent need for innovative, out-of-the-box solutions to enhance the credibility and effectiveness of sustainability assurance.

Contemporary digital technologies offer a promising pathway to address these challenges (see Table 1 for details). For instance, advanced digital tools such as blockchain, AI and big data analytics can revolutionize sustainability assurance practices (De Silva, Gunarathne and Kumar, 2024). The proponents of blockchain argue that it ensures data immutability, fostering trust in disclosures. Further AI-driven analytics identify patterns, inconsistencies and risks aiding assurance providers in achieving objectives. Meanwhile, the IoT automates the capture of environmental and operational metrics, enhancing data accuracy and reliability (Nižetić, Šolić, Gonzalez-De and Patrono, 2020) – contributing to the efficiency of assurance procedures and materiality assessment during the assurance engagement. The use of these technologies streamlines the assurance process but also improves the overall quality and credibility of sustainability reports, aligning stakeholder expectations with evolving ESG commitments. Together, they present an opportunity to overcome the dual challenges of assurance quality and the complexities associated with assurance procedures.

However, the literature capturing the implication of such technologies on sustainability assurance remains very limited. For example, Bakarich *et al.* (2020) analysed corporate cases in the USA and Canada, finding that companies adopting blockchain technology produce higher-quality information in their sustainability reports. This enhanced data integrity enables assurance professionals to verify information with minimal effort, thereby significantly improving assurance quality. De Silva *et al.* (2024), in their systematic literature review, echoed similar findings, suggesting that IoT enhances the accuracy of ESG reporting while AI improves life cycle assessments and reporting authenticity – both of which have implications for sustainability assurance.

Nevertheless, these technologies also carry risks that could undermine reporting quality and lead to suboptimal assurance. For instance, De Villiers *et al.* (2022) raised concerns about AI tools like ChatGPT, which can generate human-like content, potentially contributing to “greenwashed” reports that misrepresent sustainability efforts. Similarly, Crawford, Kastrup, and Monazzam (2023) explored the adoption of advanced digital technologies by audit firms to improve audit quality, focusing on their application to sustainability assurance in Sweden. Their study, drawing on data from multiple sources – including interviews with investors, auditors and regulators, as well as archival evidence – indicates that the adoption of digital technology has not proven effective in enhancing assurance engagements within the Swedish context. Such findings highlight both the potential and the challenges of leveraging digital technologies in sustainability assurance, offering a glimpse into how their role may evolve in the future.

3. Papers in this special issue

The papers in the special issue cover many of the approaches that can be taken in the digital technology and sustainability accounting, reporting and assurance space. The papers as summarized in Table 2 reflect diverse foci and a range of countries and approaches. Papers range from using 3D game engine technology in visualizing sustainability data, green accounting and artificial intelligence, application of digital knowledge in sustainability accounting, reporting and assurance, corporate digital transformation and audit signals, social media usage for sustainability reporting, literature review on digitalizing of a

Table 2. Technology coverage, key focus and findings of SI papers and future research directions

Technology type	Relevant SI paper(s)	Key focus and findings	Future research directions	
			Sustainability accounting	Sustainability reporting
Cloud computing	Valentinetti and Rea (2024)	Identify technological and environmental enablers (e.g., compatibility, stakeholder pressure) and organizational constraints (e.g., training needs) influencing digital adoption in sustainability accounting and reporting	- Examine how cloud computing supports Industry 4.0 for real-time environmental management accounting - Evaluate challenges in adopting cloud computing for ESG data processing across various organizational contexts	- Investigate cross-border sustainability reporting consistency using cloud-based platforms - Examine how cloud computing enhances real-time ESG disclosures - Explore cybersecurity risks in cloud-based sustainability assurance systems - Examine whether and how AI-powered cloud models enhance sustainability risk assessments
Big data analytics	De Silva et al. (2024) , Anarbaeva and Garst (2024)	This literature review study examines AI, blockchain, IoT and big data in sustainability accounting, reporting and assurance, highlighting their potential to enhance accountability and transparency through automation and data quality improvements. Anarbaeva and Garst (2024) examine digital	- Examine the role of big data in predictive sustainability accounting and materiality assessments - Examine how institutional logics, including financial institutions,	- Examine the extent to which big data enhances the accuracy, consistency and accessibility of ESG disclosures, mitigating information asymmetry - How do big data tools enhance the effectiveness of risk-based sustainability assurance frameworks? - What are the cost-benefit trade-offs associated with integrating big

(continued)

Table 2. Continued

Technology type	Relevant SI paper(s)	Key focus and findings	Sustainability accounting	Future research directions	
				Sustainability reporting	Sustainability assurance
Artificial intelligence (AI) and NLP	Khan and Gupta (2024), Farooq <i>et al.</i> (2024a, 2024b), Huang <i>et al.</i> (2024)	data accessibility, ownership and accountability in value chains, proposing a framework for measuring societal impact	cultural factors and regulatory frameworks, influence big data adoption in sustainability practices	- Analyse the effectiveness of real-time ESG analytics in identifying sustainability risks and informing evidence-based decision-making - Investigate how digital data ownership impacts ESG reporting quality	data into sustainability assurance engagements?
				- How can AI be leveraged for sustainability litigation risk assessment, particularly in predicting greenwashing lawsuits? - To what extent can AI automate sustainability accounting in	- How does AI-based fraud detection contribute to the effectiveness of sustainability assurance? - What impact do AI-driven audits have on auditor skepticism in sustainability assurance? (continued)

Table 2. Continued

Technology type	Relevant SI paper(s)	Key focus and findings	Sustainability accounting	Future research directions	
				Sustainability reporting	Sustainability assurance
Internet of Things (IoT)	De Silva <i>et al.</i> (2024) Thuy (2025)	Huang <i>et al.</i> (2024) examine corporate digital transformation and find that companies investing in AI and automation are more likely to receive standard unqualified audit opinions, signaling improved financial reporting credibility	complex, multi-tiered supply chains? • How can AI-driven models optimize environmental impact assessment and carbon footprint calculations?	sustainability challenges? • Examine the role of AI in standardizing sustainability reporting across industries	• Whether and how whether AI-driven digital transformation improves assurance credibility in financial and ESG reporting
		De Silva <i>et al.</i> (2024) find that IoT enables real-time ESG data collection, improving reporting automation and data accuracy. Thuy (2025) finds that green process innovation (GSI) and green product innovation (GPI) mediate the relationship between environmental strategy and environmental performance (ENP)	• Examine whether IoT-driven sustainability tracking leads to real behavioural change in firms (e.g. emission reductions) • Explore IoT-driven cost-saving mechanisms in sustainability accounting	• How does IoT enhance industry-specific environmental reporting standards? • What is the role of real-time IoT reporting in improving sustainability monitoring and compliance?	• What are the technical, operational, and regulatory challenges in implementing IoT-enabled real-time sustainability audits, and how can they be addressed? • How do regulatory frameworks impact the integrity, (continued)

Table 2. Continued

Technology type	Relevant SI paper(s)	Key focus and findings	Sustainability accounting	Future research directions	
				Sustainability reporting	Sustainability assurance
Blockchain technology	De Silva <i>et al.</i> (2024)	Highlights blockchain as a tool for secure ESG data collection and assurance, improving transparency and accountability	To what extent do blockchain-based sustainability accounting systems mitigate corporate greenwashing, and what factors influence their effectiveness?	How do investors and regulators perceive the credibility and reliability of blockchain-verified ESG reports compared to traditional disclosures?	- reliability, and standardization of IoT-collected ESG data for sustainability reporting and assurance? - Examine whether green technology can be integrated into third-party sustainability assurance standards
				How can blockchain technology improve traceability in sustainability audits and assurance engagements, and what are the key barriers to its adoption?	(continued)

Table 2. Continued

Technology type	Relevant SI paper(s)	Key focus and findings	Future research directions		
			Sustainability accounting	Sustainability reporting	Sustainability assurance
3D Game Engine Technology and VR	Muhammad and Hazelton (2024)	The study indicates that although the 3D environments improve engagement with sustainability data but reduce navigation speed and data clarity	How can virtual reality be leveraged to enhance the training of sustainability accountants in understanding and applying complex ESG metrics?	How does interactive virtual reality-based sustainability reporting impact regulatory compliance and stakeholder engagement?	What is the potential of virtual reality in real-time sustainability auditing, and how effective is it for remote ESG compliance inspections?
			Examine whether sustainability accountants use social media as an ESG data source, and how it impacts reporting accuracy and credibility	To what extent do social media-driven ESG disclosures influence corporate sustainability performance, and what factors mediate this relationship?	Should social media-based ESG claims be subjected to external assurance, and what frameworks or criteria would ensure their credibility and reliability?
Social media disclosure (Facebook, X, LinkedIn)	Hogarth et al. (2024)	This paper finds that sustainability-related social media content is underutilized, and when used, focuses more on social rather than environmental issues			
			How can RPA be leveraged to automate ESG tax compliance for multinational corporations, and	What are the key barriers to workforce adoption of RPA in sustainability reporting, and	To what extent do RPA-driven sustainability audits enhance efficiency, and what are the
Robotic process automation (RPA)	–	–			

(continued)

Table 2. Continued

Technology type	Relevant SI paper(s)	Key focus and findings	Sustainability accounting	Future research directions	
				Sustainability reporting	Sustainability assurance
Data visualization and dashboards	Muhammad and Hazelton (2024)	Finds that 3D data visualization improves engagement but may reduce clarity in sustainability reporting	what are the associated challenges and benefits?	what strategies can organizations implement to overcome resistance and enhance integration?	potential trade-offs in reducing human judgment and professional skepticism?
				How do different data visualization formats impact the psychological perception and readability of ESG reports?	- What mechanisms can sustainability assurance providers implement to detect and mitigate the effects of misleading sustainability visualizations? - How should sustainability assurance frameworks address misleading visualizations to enhance stakeholder trust?

Source(s): Authors' own work

technology providers' products and services, challenges in measuring social impact, AI adoption on ESG disclosure, environmental management accounting through green technology and digital technologies and sustainability accounting, reporting and assurance.

The use of 3D game engine technology for disseminating sustainability information is explored by [Muhammad and Hazelton \(2024\)](#). They find that visualization of sustainability accounting data and information through a "3D virtual environment offered enhanced engagement and contextual understanding but reduced navigation speed and data clarity. Participants were enthusiastic about the potential of a museum-like experience, incorporating interactivity and community, but felt that the appropriate audience was more likely to be novices than experts". Their study offers a novel contribution to the data visualization literature within the context of sustainability accounting, reporting and assurance. The aim of data visualization is improving information communication effectiveness. The aim is to improve users' understanding of data and information.

In their multicountry study, [Khan and Gupta \(2024\)](#) analyse the moderating role of country-level AI infrastructure and accounting education quality on the relationship between green accounting practices and corporate financial performance. They observe a positive relationship between green accounting practices and corporate financial performance. Furthermore, they find that country-level AI infrastructure positively and country-level accountancy education quality negatively moderate this relationship. Integrating AI enhances the data processing, predictive analytics and decision-making, improving green accounting effectiveness.

[De Silva et al. \(2024\)](#) review the literature to examine the impact of digital technologies on sustainability accounting, reporting and assurance. The review reveals key themes including AI, blockchain, IoT and big data. Researchers are optimistic that these technologies can play a positive role in enhancing corporate accountability and transparency through automation and improved data quality. However, these papers have been published primarily in nonaccounting journals. They find that the IoT can be used to facilitate the automation of sustainability reporting processes with implications for the quality of disclosures. Furthermore, there is opportunity for digital technologies to be leveraged in assisting with the integrating governance mechanisms and sustainability accounting and reporting. Finally, digital technologies can be leveraged to support the e-taxonomies. The use of blockchain in data collection is also noted with implication for data collection and subsequent external assurance.

[Huang et al. \(2024\)](#) examine the impact of corporate digital transformation on the audit opinions of auditors in A-share nonfinancial listed companies in China. Their study fills a gap in the literature by investigating the impact of digital transformation on business performance, especially in the Chinese context, where digital transformation is rapidly progressing. [Huang et al. \(2024\)](#) study reveal that digital transformation leads to standard unqualified audit opinions, meaning that companies that invest more in digital technologies, and capabilities has more tendency to receive standard unqualified audit opinions, which signify the reliability and credibility of their financial reporting.

[Hogarth, Lodhia, Kaur and Stone \(2024\)](#) examine the extent, nature and communication potential of companies' use of three popular social media platforms (Facebook, X and LinkedIn) to report on sustainability. The media richness framework was used to measure the communication potential of the social media platforms for sustainability communication. The results indicate that the extent of sustainability posts on social media represented less than 20% of total social media posts. The nature of posts by the Top 50 ASX companies was higher on social issues than on environmental issues, which is contradictory to many previous studies. The study also found that while the social media platforms afforded high

level of media richness, most companies failed to exploit the platforms' full potential to disseminate sustainability information. Hogarth *et al.* (2024) study extends existing literature, particularly, the Lodhia *et al.* (2020) study, which has primarily focused on examining sustainability content in the media with limited exploration of the communication potential of social media platforms to communicate sustainability content.

Valentinetti and Rea (2024) examine the factors related to the emergence of digital accounting, reporting and disclosure of sustainability corporate information. Following the Technology-Organization-Environment framework, the authors review and classify the factors identified by current academic literature that may enable or constrain the use of digital solutions for collecting, reporting and disclosing corporate nonfinancial information. The results show an increasing interest in the digitalization of corporate sustainability accounting and reporting and highlight some related drivers with a predominance of technological and environmental enablers (e.g. compatibility, stakeholders' pressure) along with organizational constraints (e.g. need for technical and training programmes) influencing the ongoing adoption of new technologies for both internal and interorganizational purposes.

Anarbaeva and Garst (2024) in the article entitled "Accounting for downstream value chain: examining the accountability for social impact of digitalisation" examine the tensions between digital data accessibility, quality and accountability. They create a framework of attributes which measures companies' societal impact in their value chains. The research boundaries are set around the societal impact arising from the technology providers' primary activities – development and implementation of digital technologies and information systems – downstream value chain. The authors conducted participatory action research in a single case, a technology provider. The five attributes: accuracy, data availability, digital data ownership, relevancy of information and accountability are good indicators that need to be possessed to measure uncaptured value in the downstream value chain.

Farooq *et al.* (2024a, 2024b) in the article entitled "AI adoption, ESG disclosure quality and sustainability committee heterogeneity: Evidence from Chinese companies" investigate the adoption of AI influence on the quality of corporate sustainability disclosures. The authors found that AI adoption enhances overall sustainability disclosure quality and pillar specific ESG disclosure quality. This positive effect is further amplified by the presence of sustainability committees. Examining the heterogeneous impact of these committees revealed stronger association between specific committee focus (sustainability committee specialization) and relevant ESG pillar disclosure quality (except for governance), suggesting targeted committees can optimize disclosure outcomes. The authors find evidence to support the creation of sustainability committees, as a key corporate governance mechanism to drive corporate sustainability reporting.

Thuy (2025) in the article entitled "unlocking environmental management accounting and environmental performance: a mediated moderation model through green technology innovation and environmental strategy" examines the intricate connections between environmental management accounting and environmental performance through the mediating and moderating roles of green technology innovations, including green process innovation (GSI) and green product innovation (GPI) and environmental strategy (EST). The study uses a mediated moderation model with the lenses of legitimacy theory and the natural resource-based-view to explore how these constructs interact. It then offers insights into how organizations can improve environmental performance (ENP) by integrating environmental management accounting (EMA), green technology innovation and EST. The study uses a quantitative approach and accumulates data from a survey of 325 senior managers and chief accountants of manufacturing firms in Vietnam. The findings indicate a significant positive correlation between EMA, GSI, GPI and ENP. In addition, the findings reveal that the two

forms of green technology innovation, namely, GSI and GPI, mediate the relationship between EMA and ENP. Furthermore, [Table 1](#) in the earlier section of the paper also identifies research findings in prior studies and the potential gap that still persists in the literature. This includes developing frameworks to use blockchains for tracking sustainability related data, how companies are using social media platforms to disseminate sustainability information amongst others.

4. Future research directions

Future research may aim to establish how Industry 4.0 might facilitate more accurate, high quality, real time environmental management accounting and external environmental reporting in relevant sectors, company sizes, across different management roles and collaborative settings, as well as in intraorganizational settings such as supply chains. There is a need for organizations to embrace Industry 4.0 and to facilitate this, research has to go hand in hand with practical developments and map potential uses of these technologies. [Table 2](#) outlines the potential future research possibilities in relation to use various technologies in sustainability accounting, reporting and assurance.

Researchers can also examine barriers to adoption of digital technologies as well how sustainability reporting managers (as well as sustainability assurance practitioners) are overcoming these barriers to adoption. These barriers (financial and cultural) may exist at an individual, organization and national level. An understanding of these barriers and mechanisms to resolve them will enable the full potential of digital technologies to be used and enable organizations to be aware of existing challenges.

The use of alternative theories and methodologies can offer interesting new insights on the impact of digital technologies on evolution of the field. Existing research in sustainability accounting and reporting is limited to theoretical insights that emphasize information content. As digital technologies play a critical role in sustainability processes, there is a need for theorization of these technologies and their capabilities. For instance, Hogarth and colleagues in this special issue highlight the utility of communication theories such as media richness framework to go beyond information content and assess communication potential of social media disclosure. Similarly, new developments in research methodologies that seek to address the role of digital technologies are needed to keep pace with the rapid developments in digital technologies.

While the benefits of digital technologies in sustainability accounting, reporting and assurance have been discussed, researchers can also compare the proclaimed benefits against the actual benefits from those who have experience in using these technologies. There is also a need for further investigation into how emerging technologies can be leveraged to achieve sustainability goals.

Despite the growing importance of digitalization in sustainability assurance, there is a noticeable gap in research capturing the nuances of how these technologies influence various facets of assurance practice. Current studies offer limited insights, leaving much to explore regarding their impact on assurance processes and outcomes. For instance, while digital technologies are frequently lauded for enhancing assurance quality (see, [De Silva et al., 2024](#)), there remains a lack of longitudinal empirical evidence to substantiate these claims. Similarly, while big data tools undoubtedly expand the scope and engagement of assurance providers (see, [Salijeni, Samsonova-Taddei and Turley, 2021](#)), the cost-benefit dynamics of adopting such tools remain largely unexamined.

Furthermore, evidence suggests that digital technologies enhance the readability of reports ([Ante, 2022](#); [Shimamura et al., 2025](#)), yet this analysis has not been fully extended to sustainability assurance reports. This represents a critical gap in understanding how such

technologies could improve the clarity and accessibility of these reports. Advanced tools like data visualization software and dashboards have the potential to transform complex ESG metrics into intuitive formats – such as graphs, charts and infographics – making them more accessible to a diverse range of stakeholders. Similarly, AI can further enhance report clarity by summarizing key findings and structuring content through natural language processing. However, these benefits come with risks; the same tools could be misused for greenwashing, allowing organizations to present misleading sustainability claims (De Villiers, Dimes and Molinari, 2024). We believe this dual potential – the bright and dark sides of technology adoption – highlights the pressing need for a more in-depth examination of how digital technologies can be responsibly harnessed to enhance the credibility and transparency of sustainability assurance reports.

5. Conclusion

This article examined the role of digital technologies in sustainability accounting, reporting and assurance. It also discussed the articles of the special issue and identified future research directions. This article is a theoretical piece which draws upon existing work to highlight the use of digital technologies in sustainability accounting, reporting and assurance. Sustainability is critical for our survival and emphasis on it has mirrored the rapid development of digital technologies. The unique aspect of our work is that we focus on the entire accounting, reporting and assurance process and highlight how digital technologies can be used for sustainability accounting, reporting and assurance with an emphasis on both the enablers and the barriers.

In conclusion, we would highlight that we have only touched the surface in relation to the use of digital technologies in sustainability accounting, reporting and assurance in relation to this special issue. There is much more research that is required given the phenomenal developments in digital technologies and the increasing urgency associated with sustainability issues. We hope that this special issue will encourage further research in an increasing critical area of research.

This paper makes theoretical and practical contributions. At the theoretical level, we have expanded the limited literature on the use of digital technologies in sustainability accounting, reporting and assurance through our work as well as via the discussion of the significant work undertaken by the authors of this special issue. We also discuss some further areas of inquiry that could be undertaken in future work. From a practical perspective, our paper as well as the papers in this special issue provide extensive guidance to practitioners on the use of digital technologies in sustainability accounting, reporting and assurance.

Notes

1. For example, the new European Union Corporate Sustainability Reporting Directive (EU CSRD) requires reporters to secure limited assurance over their sustainability reports. However, the sustainability assurance market remains open to a range of accounting and non-accounting sustainability assurance providers.
2. www.ifac.org/knowledge-gateway/discussion/iaasb-approves-landmark-issa-5000-standard-new-era-sustainability-assurance

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